

Pressure Vessel Visual Inspection Procedure

1. Purpose

This Standard Operating Procedure (SOP) establishes a method for visually inspecting pressure vessels in a petroleum refinery. Its purpose is to identify visible deterioration, damage, leakage indicators, or unsafe conditions early, support safe operation, and ensure findings are consistently recorded and escalated. Visual inspection does not replace engineering assessment, thickness measurement, non-destructive examination.

2. Scope

This procedure applies to accessible external surfaces of refinery pressure vessels, including shells, heads, nozzles, flanges, supports, welds, insulation. Inspections shall be performed by trained and authorized personnel under refinery permit-to-work, isolation, gas-testing, requirements. Internal inspection is outside this visual procedure unless authorized by an approved inspection plan.

3. Required PPE and Safety Controls

3.1 Wear flame-resistant clothing, safety helmet, safety glasses, safety footwear, suitable gloves, and hearing protection.

3.2 Use chemical-resistant PPE when the hazard assessment requires it.

3.3 Before inspection, confirm equipment status, isolation, depressurization, draining, purging, gas testing, and permit requirements.

3.4 Use scaffolds, ladders, platforms, lighting, and fall protection. Never enter confined spaces or approach hazardous areas without authorization and required controls.

4. Inspection Checklist

4.1 Shell and Heads: Examine accessible surfaces for dents, bulging, distortion, cracks, mechanical damage, staining, leaks, unusual deposits, or signs of overheating.

4.2 Nozzles and Flanges: Inspect nozzle necks, reinforcement areas, flange faces, bolts, nuts, and visible gaskets for cracking, corrosion, distortion, leakage, loose hardware, or abnormal deposits.

4.3 Welds: Examine longitudinal, circumferential, nozzle, attachment, and repair weld areas for visible cracks, undercut, porosity indications, corrosion grooves, weld distortion, or separation. Pay special attention to heat-affected areas.

4.4 Corrosion: Look for rust, pitting, scale, thinning indications, blistering, coating failure, galvanic attack, corrosion under supports, and evidence of process leakage. Record locations and approximate dimensions. Do not scrape aggressively or disturb protective coatings without authorization.

4.5 Insulation and Cladding: Inspect insulation, weatherproofing, bands, seams, penetrations, and jacketing for wet spots, staining, bulges, damage, missing sections, or openings. Suspected corrosion under insulation shall be escalated for authorized examination.

4.6 Supports and Attachments: Check skirts, saddles, legs, clips, platforms, ladders, earthing connections, and visible attachments for corrosion, cracking, deformation, looseness, or damaged connections.

4.7 Identification: Verify vessel identification, nameplate visibility, inspection status, and abnormal operating indicators against approved records.

5. Acceptance and Rejection Criteria

5.1 Accept when inspected areas show no unacceptable cracks, active leakage, significant deformation, unsafe attachments, or deterioration beyond approved limits, and protective systems remain serviceable.

5.2 Reject or place the vessel in controlled hold when active leakage, through-wall damage, significant cracking, severe local corrosion, major distortion, damaged pressure-boundary welds, or other conditions indicate possible loss of containment or structural integrity.

5.3 Visual findings shall not be declared acceptable solely by inspector judgment when engineering limits, thickness data, codes, or statutory requirements apply. Refer uncertain conditions for engineering evaluation.

6. Documentation

Record vessel identification, date, inspector name, operating status, inspection location, access limitations, observed defects, photographs, approximate dimensions, and immediate actions. Reference drawings, previous reports, inspection plans, and relevant work orders. Attach photographs. Enter findings into the refinery inspection or maintenance system and retain records according to site requirements.

7. Escalation Procedure

7.1 Stop or restrict inspection if unsafe conditions arise.

7.2 Immediately report active leakage, suspected pressure-boundary failure, severe cracking, major deformation, or critical defects to the operations supervisor and inspection engineer.

7.3 Establish required operating restriction, isolation, barricading, monitoring, or shutdown through authorized refinery personnel.

7.4 Inspection engineering shall determine additional examination, such as thickness measurement or appropriate non-destructive testing, and establish disposition.

7.5 Close the finding only after authorized review, corrective action, and documented acceptance. Follow refinery emergency procedures whenever personnel safety or containment is threatened.